

ADDIS ABABA BOLE INTL, Ethiopia

WMO: 634500

Lat: 8.978N

Lon: 38.799E

Elev: 2326

StdP: 76.33

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	8.9	-7.0	2.8	17.2	-4.0	3.6	16.1	7.7	16.3	6.6	17.3	1.9	330	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.2	26.2	13.8	25.5	13.9	24.9	13.9	17.9	22.6	17.0	21.8	16.3	21.2	5.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	15.4	21.2	15.2	14.4	19.7	14.6	13.8	18.8	61.1	22.7	57.5	21.7	55.2	21.6	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	16.5	17.9	19.0	18.7	18.8	18.0	16.6	16.3	16.8	17.1	16.7	15.9		
	DBStd	1.63	1.38	1.39	1.51	1.38	1.35	1.33	1.06	0.94	1.06	1.14	1.23	1.30		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	464	57	23	10	12	11	22	55	63	48	40	50	75		
	CDD10.0	2685	203	220	279	262	273	239	204	196	203	221	201	184		
	CDD18.3	109	2	9	31	24	26	12	1	0	1	2	1	1		
	CDH23.3	656	32	90	182	122	140	64	1	0	2	11	7	4		
	CDH26.7	12	1	1	6	3	2	1	0	0	0	0	0	0		
(14) Wind	WSAvg	3.5	3.8	4.1	3.9	3.5	3.6	3.1	2.9	2.7	2.7	4.0	3.9	3.9		
(15) Precipitation	PrecAvg	1197	18	36	65	86	82	128	255	280	169	39	10	10		
	PrecMax	1556	128	168	233	229	325	258	397	426	386	273	62	65		
	PrecMin	904	0	0	0	6	0	38	181	124	56	0	0	0		
	PrecStd	155	24	39	54	52	63	45	45	67	60	49	17	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	26.2	27.2	27.0	26.9	26.2	23.0	22.2	23.6	24.2	24.1	24.1		
		MCWB	12.7	13.1	13.5	14.2	14.0	14.1	16.0	16.0	14.8	14.1	14.3	13.2		
	2%	DB	24.1	25.2	26.2	25.8	25.9	25.0	21.6	21.1	22.3	23.5	23.1	23.1		
		MCWB	12.5	13.3	13.4	14.3	14.3	14.5	15.3	15.7	15.0	13.9	14.1	13.1		
	5%	DB	23.2	24.4	25.2	24.8	25.0	24.0	20.7	20.1	21.4	22.8	22.5	22.2		
		MCWB	12.3	13.2	13.5	14.5	14.3	14.5	15.2	15.2	15.0	13.7	13.9	12.7		
	10%	DB	22.2	23.7	24.1	23.8	24.0	22.7	19.8	19.2	20.5	22.0	21.8	21.6		
		MCWB	12.1	13.0	13.3	14.6	14.4	14.9	15.0	14.9	14.9	13.1	13.6	12.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	18.6	19.0	18.4	18.9	18.0	17.1	17.2	17.3	17.0	17.1	16.6		
		MCDB	22.7	23.6	23.9	23.1	24.1	22.3	20.7	20.5	21.0	22.2	21.6	21.8		
	2%	WB	15.2	16.6	16.6	17.0	17.5	16.8	16.1	16.2	16.2	15.7	15.8	15.0		
		MCDB	21.9	22.7	21.7	22.1	22.7	21.8	20.2	19.7	20.3	21.3	21.2	21.2		
	5%	WB	14.5	15.1	15.3	16.1	16.3	15.9	15.5	15.6	15.7	14.8	15.1	14.2		
		MCDB	21.0	21.4	21.6	21.8	21.7	21.2	19.5	19.1	19.8	20.6	20.8	20.5		
	10%	WB	13.7	14.2	14.6	15.4	15.5	15.3	15.0	15.1	15.2	14.3	14.4	13.4		
		MCDB	20.1	20.7	20.9	21.2	21.0	20.6	18.8	18.5	19.2	19.8	20.1	19.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	12.7	11.2	10.4	10.2	9.8	7.2	7.1	8.2	10.6	11.3	12.5		
		MCDBR	14.1	13.5	12.2	12.1	12.1	11.8	8.5	8.2	9.4	11.7	12.0	13.2		
	5% WB	MCWBR	6.8	6.3	5.1	5.1	5.5	5.4	4.4	4.4	5.0	6.0	6.5	6.6		
		MCWBR	11.8	11.8	10.8	9.8	10.1	10.0	7.9	7.7	8.4	10.1	10.7	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.333	0.361	0.368	0.405	0.505	0.497	0.432	0.412	0.337	0.306	0.301		
	taud		2.513	2.461	2.407	2.419	2.268	1.934	1.981	2.208	2.248	2.456	2.532	2.537		
	Ebn at Noon		1002	987	957	937	885	789	799	867	895	970	1001	1007		
	Edh at Noon		106	115	123	120	137	188	181	147	142	114	103	101		
(44) All-Sky Solar Radiation	RadAvg		6.13	6.63	6.60	6.24	6.19	5.76	5.04	4.93	5.51	6.20	6.11	6.02		
	RadStd		0.41	0.46	0.53	0.38	0.32	0.23	0.37	0.33	0.28	0.33	0.25	0.51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page